Corporate Bankruptcy Prediction

UE24CS352A - Machine Learning | Project write-up

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Abstract. This project predicts bankruptcy within three years using financial ratios from Polish companies. Six classifiers are compared using development-set cross-validation. Histogram gradient boosting is selected, and a screening threshold is chosen from out-of-fold predictions. On the independent test set, the model detects 84 of 99 bankrupt statements, with 34.3% precision and 0.750 average precision. The remaining false alarms and missed cases define the main limitation.

1. Problem and objective

The task is binary classification: label 1 indicates bankruptcy during the three-year horizon and label 0 indicates no bankruptcy in that horizon. Because positive cases are rare, high accuracy alone is insufficient. The objective is to compare ranking quality and evaluate a threshold that places greater emphasis on recall.

2. Dataset and data preparation

The project uses UCI's **3year.arff**, containing 10,503 financial statements, 64 financial ratios and 495 positive labels [1]. It removes 87 exact duplicate rows before splitting, leaving **10,416 statements and 493 positives (4.73%)**. No identical feature vectors with conflicting labels were found. Missing values remain available to the model pipeline.

For logistic regression, naive Bayes and SVM, preprocessing uses median imputation with missingness indicators, an inverse-hyperbolic-sine transform and standard scaling. Forest models use imputation and indicators. Histogram gradient boosting handles missing values natively. Each fitted transformation is learned inside its training fold.

3. Experimental design

A stratified split with seed 42 reserves **2,084 rows (99 positives)** for final testing. The remaining **8,332 rows (394 positives)** are evaluated with identical three-fold stratified splits. Each algorithm uses one fixed configuration. The highest mean cross-validation average precision selects the model. The selected model's out-of-fold scores determine a threshold that maximizes F2; neither choice uses test labels [2].

4. Cross-validation comparison

Model	Mean AP	Fold SD	Mean ROC AUC
Histogram gradient boosting	0.619	0.037	0.914
Random forest	0.434	0.037	0.868
RBF support vector machine	0.422	0.053	0.873
Extra trees	0.383	0.050	0.852
Logistic regression	0.297	0.008	0.871
Gaussian naive Bayes	0.087	0.034	0.676

AP = average precision. Fold standard deviation describes the three development folds; it is not a confidence interval. Histogram boosting uses 250 iterations, learning rate 0.07, up to 15 leaves, L2 regularization 1, and balanced class weights.

5. Independent test results

The chosen threshold is **0.1980**. After fitting on the development set, the frozen model achieves **84.8% recall, 34.3% precision, 0.750 AP and 0.968 ROC AUC**. Its confusion matrix is TN=1,824, FP=161, FN=15, TP=84. The all-negative baseline has 95.25% accuracy and zero recall.

Threshold rule	Recall	Precision	F2	FP / FN
OOF F2 choice (0.198)	84.8%	34.3%	0.655	161 / 15
Default 0.500	68.7%	65.4%	0.680	36 / 31

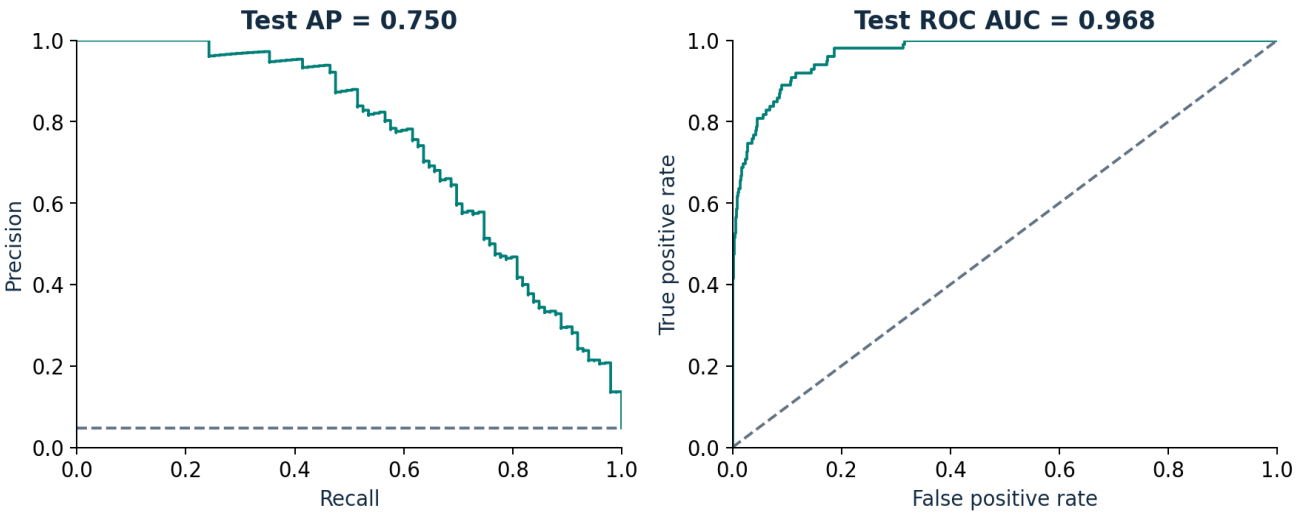


Figure 1. Held-out precision-recall and ROC curves. The dashed precision baseline is test prevalence. The chosen threshold gains 16 detections and 125 false alarms relative to 0.5. Default 0.5 has higher test F2 in this split; the original threshold is retained to avoid selection on the test set.

Using 500 stratified bootstrap resamples of test predictions, approximate 95% intervals are 77.8-91.9% for recall, 31.1-38.7% for precision, and 0.683-0.820 for AP. These intervals condition on the fixed model and split and do not include retraining or distribution-shift uncertainty.

6. Implementation, interpretation and conclusion

The repository provides modular training and inference, a saved model and checksum, exact split and prediction records, an executed notebook, automated tests, and a local Flask demo with CSV input. The demo includes correctly classified examples and real errors. Permutation importance identifies X27 (operating profit / financial expenses), X34 (operating expenses / total liabilities), and X46 (liquidity ratio) as influential, without establishing causality.

The experiment supports a useful ranking signal in this dataset, while 161 false alarms and 15 missed cases remain. Scores are uncalibrated. Historical Polish data may not transfer to new settings; missing company IDs prevent verified company-disjoint validation, and a random split is not a temporal forecasting test. Future work should add group/date information, assess threshold stability, and validate calibration on independent data.

References

[1] S. Tomczak, *Polish Companies Bankruptcy*, UCI, 2016. doi:10.24432/C5F600. CC BY 4.0.
[2] Scikit-learn documentation: model evaluation and cross-validation, scikit-learn.org.
[3] N. A. Patil, *Corporate Bankruptcy Prediction*, supplied reference report; codeatri/Bankruptcy_Prediction, GitHub. Background only; this implementation and reported experiment are independent.